

An agent-executed, preregistered reproduction pipeline for BAO
cosmology with kernel-checked model identities:
three preregistered DESI/SDSS reproductions and one earlier
reproduction as one methodology study

Xavier Callens (SocrateAI Lab)

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Abstract

We describe and audit a reproduction pipeline for baryon acoustic oscillation (BAO) cosmology in which the analysis is executed by language-model agents, every number is produced by a script in the repository, targets and tolerances are preregistered, the algebraic skeleton of each closed-form model is stated and kernel-checked in Lean 4 (the fits and numerics are not machine-checked; for the H_0 run only the secondary fitting-formula r_d has a Lean skeleton, the primary CAMB r_d has none), claims are filed in a tier-capped ledger, and an adversarial model referee re-runs each preregistered analysis before a fix round. The pipeline was applied to four problems. (i) DESI DR1 flat Λ CDM (not preregistered): $\Omega_m = 0.2939$, $hr_d = 101.94$ Mpc (χ^2 minimum), matching DESI's printed best fit (0.294, 101.94 Mpc); our $\chi^2_{\min}$ is higher than DESI's 12.66 by 0.081, an offset we could not explain. (ii) DESI DR2: $\Omega_m = 0.29781 \pm 0.00856$, $hr_d = 101.530 \pm 0.732$ Mpc, and w CDM $w = -0.9164 \pm 0.0787$; the largest of the four |pulls| is 0.037, and 14 of 14 preregistered criteria hold. (iii) BAO+BBN inverse distance ladder: $H_0 = 68.545 \pm 0.594 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DR2, exact CAMB sound horizon, verdict PASS) against 68.51 ± 0.58 ; the DR1 check gives 68.697 ± 0.809 against 68.53 ± 0.80 , an offset of 0.167 that exceeds the preregistered strict window of 0.15 (verdict PARTIAL in the recorded run). Re-executing the same frozen script with the same seeds under the project's other Python environment gives $|\Delta H_0| = 0.1455$, i.e. PASS; the two runs differ by 1.5 Monte Carlo errors, so this label depends on the numerical environment and the strict boundary is not resolved. The preregistered BAO-only gate tolerance admits H_0 offsets larger than the strict window, so PASS versus PARTIAL does not isolate the BBN/ r_d step. (iv) SDSS/eBOSS vs DESI DR2: SDSS $\Omega_m = 0.2987 \pm 0.0164$ against 0.299 ± 0.016 . The cosmological values are reproductions, not new measurements. The pipeline also yields two simple functions of public inputs that we did not find quoted in the fetched DESI text: the DR1→DR2 shift $\Delta\Omega_m = 0.00283$, which is 0.237 times the nested-sample uncertainty (0.46 under a perfect-correlation bracket); and, for BAO-only (Ω_m, hr_d) with DESI's eq. (18) statistic, an SDSS–DESI DR2 tension of $N_\sigma = 0.877$ (PTE 0.381) under the independent-survey assumption. Because the surveys overlap, N_σ is a model-dependent lower bound (and the PTE an upper bound): an illustrative cross-covariance with $\rho = 0.57$ raises it to 1.34. We report the verification artifacts as well: 28 Lean theorems (7 of them short identities or positivity statements), all with whitelist-only axioms in the pinned build and 22 of them re-checked against a fully built Mathlib (same Lean binary and Mathlib commit, so this tests independence from the partial build, not from the toolchain); a hash-locked, unread first fit that is reported next to the amended analysis; a wrong-convention control that the data cannot detect ($\Delta\chi^2 = -5.2 \times 10^{-5}$) and that the external target only marginally catches; and a port to a Rust reimplement of CVOID (rusty-SUNDIALS, not the LLNL library) that reproduces the DR2 fit and exposed defects in that reimplement. Every audit of a Lean

statement was done by a model of the same family as the analysis agent. No person has reviewed any component.

AI-use statement. The analyses, code, Lean files, ledgers, referee reports, fixes and this manuscript were produced by Claude agents running in Claude Code under a workflow that the author designed and launched. The author chose the problems and the rules (preregistration, controls, Lean gate, ledger, referee). The only record of model tiers is an orchestrator status message (timeline, 2026-09-27T15:44:19.533Z; quote sha256 prefix 9944823b85818ab0): “Cosmology workflow: running in the background. All three problems (DESI DR2 BAO, H0 from BAO plus nucleosynthesis, eBOSS vs DESI) are in the pre-registration stage. Lean and the refuting review run on Fable; fits and papers use the default model.” The identity of the “default model” is not recorded in any artifact. This synthesis was drafted by a Claude agent in the same way. The adversarial referees are therefore Claude models reviewing Claude-written work: producer and verifier differ at most in model tier, not in vendor or training lineage, and automated review verdicts are known to depend on the reviewer’s model family (Gemini and Claude reviewers agree at Spearman $\rho = 0.907$, GPT-5.4 with either at $\rho \approx 0.32$ [35]). Each of the four problems was executed once by the agent pipeline (one `fit.json` per run); the DR2 bit-for-bit regeneration of Sec. 3.6 is a fixed-seed re-execution of the same scripts, not an independent agent run, so no run-to-run statistics exist. Every number produced by these analyses is a macro written by `papers/cosmo_synthesis/gen_numbers.py` from the result JSONs (and, for the Rust cross-check and guard logs, from the files named in Sec. 3.7 and Sec. 6), and `number_manifest.json` lists the file and key behind each one. Every JSON under `results/cosmo_synthesis/` that feeds a macro is written by a script in that directory; the first version of `lean_dual_check.json` was hand-corrected and is kept, with its hash, as `lean_dual_check_v1_handedited.json` (Sec. 3.4). Rule constants (thresholds such as 10^{-3} or 50τ), equation numbers and figures quoted from other papers are typed. **Human review status:** none documented for any component. The fits, controls, Lean statements, ledgers and this text have not been reviewed by a person. Where the text says “audited” or “refereed”, the auditor or referee was a model.

1 Introduction

BAO distance ratios D_M/r_d , D_H/r_d and D_V/r_d are among the cleanest cosmological observables. In flat Λ CDM, BAO data alone constrain only Ω_m and hr_d , and adding a BBN prior on ω_b calibrates r_d and hence H_0 . DESI published DR1 [1] and DR2 [2] results, and SDSS/eBOSS published its final compilation [3]. Public likelihood files for all three are available. That makes these results good test cases for an automated reproduction pipeline: the targets are precise, the data are public, and a failure cannot be blamed on missing inputs.

This paper does not claim new cosmology. It asks whether an agent-executed pipeline can reproduce published BAO numbers under rules strict enough that a wrong result would have been caught and a lucky one would not count, and it lists what went wrong along the way. We treat four runs as one study: the earlier DESI DR1 exercise and three preregistered problems run in parallel.

2 The pipeline

Each preregistered problem passed through the same stages. The DR1 run predates the workflow: it had a literature review but no preregistration (stage 1) and no referee (stage 6).

1. **Literature and preregistration.** Every target is taken from a fetched source (arXiv id plus equation or table) and recorded in `docs/literature/*_LITERATURE_REVIEW_2026.md`. The preregistration JSON fixes data files (sha256), model, priors, estimator, targets, tolerances, controls and the planned Lean statements before any fit is run.
2. **Fit with controls.** Positive controls (noiseless and noisy mocks, predictor cross-checks), negative controls (scrambled data, wrong model, label swaps, scrambled covariance) and, after review, a subtle-bug control. Results are written only by scripts under `scripts/<run>/` to `results/<run>/`.
3. **Lean 4 formalization** of the algebraic skeleton (positivity, monotonicity, identities), with in-file `#print axioms`, a `sorry` negative control and (added in revision) an axiom-smuggling control.
4. **Elenchus ledger:** a tier-capped claim ledger with content-addressed evidence blobs [8] (the author’s own software, unpublished and not peer-reviewed).
5. **Paper** built from generated macros.
6. **Adversarial referee**, a model agent that re-runs the analysis, followed by a **fix round**.

After the runs, two cross-checks were added: a dual-environment Lean re-check (Sec. 3.4) and a Rust/CVODE port of the distance numerics (Sec. 3.7).

3 Methods

3.1 Data and model

All DESI fits use the public Gaussian BAO files, whose sha256 hashes are listed in Table 9. DR1 has 12 data points. The model is flat Λ CDM or constant- w w CDM. In the BAO-only runs, radiation is off in the primary model and enters as a sensitivity variant. For DR2 Λ CDM, adding radiation shifts Ω_m by -0.046 published σ . Fitting the primary model to a noiseless CAMB data vector with $\sum m_\nu = 0.06$ eV returns Ω_m higher than the CAMB input by 0.032σ (primary minus CAMB), so a CAMB-model analysis of the real data would sit lower by about that amount, in the same direction as the radiation variant. Applied to the DR2 Ω_m pull of 0.037 , these offsets give 0.005 (CAMB) and -0.009 (radiation); both are negligible. The massless-neutrino CAMB background agrees with our distances to 2.3×10^{-11} . The H0 run models radiation and one massive neutrino in the background, uses the DESI flat priors and the BBN prior $\omega_b = 0.02218 \pm 0.00055$, and computes r_d with CAMB. The eBOSS run fits the SDSS DR16 compilation with its non-Gaussian likelihood tables and never combines SDSS with DESI.

3.2 Preregistration and the H_0 protocol amendment

The H0 preregistration (timestamp 2026-09-27T15:53:42Z) named the Aubourg et al. eq. 16 fitting formula [5] as the primary r_d . CAMB and CLASS were installed after it was written. Amendment 1 (2026-09-27T16:04:27Z), written by the orchestrating session rather than the fit agent, made an exact CAMB r_d table primary and kept the fitting formula as the preregistered secondary. Targets, tolerances, controls and data did not change. A first fitting-formula fit had already been written (16:03:03Z) by the fit agent. The amender did not read it, hash-locked it (sha256 prefix `b7fe5608e4ae6d0b`; a local, mtime-timestamped instance

of the pre-run commitment that TOPO [26] timestamps publicly, Sec. 5) and renamed it `fit_attempt1_fittingformula_UNREAD_AT_AMENDMENT.json`. The file’s current hash still matches the amendment: yes. **Both results are reported** (Table 3, Fig. 2). The locked attempt gives $H_0 = 68.579 \pm 0.601$ (DR2) and 68.701 ± 0.815 (DR1). The re-run secondary, in the recorded environment, is identical to it (difference 0.0). That identity is expected from fixed seeds, the same code path *and the same numerical environment*: re-executed under the other interpreter on this machine (Sec. 4.3) the secondary differs from the locked file by -0.0015 (DR2) and 0.0011 (DR1) in H_0 . The comparison therefore shows same-environment reproducibility only, not independent correctness. The strict window $|\Delta H_0| \leq 0.15 \text{ km s}^{-1} \text{ Mpc}^{-1}$ was built in the preregistration as $\sqrt{0.139^2 + 0.05^2} = 0.148 \rightarrow 0.15$, from a fitting-formula systematic of $0.139 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and a Monte Carlo and numerics allowance of $0.05 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (the preregistration’s own estimate of the MC noise was “ ~ 0.02 ”). With the exact r_d the formula term no longer applies. A post-hoc re-budget keeps the MC allowance and adds a CAMB-table interpolation term of 0.0044 , giving 0.050 ; it was *not* used: the preregistered rule decides the verdict. The preregistered prior probability of a strict PASS was 0.55 .

The preregistered tolerances are not mutually coherent. The BAO-only gates G1/G2 accept $|hr_d - \text{target}| \leq 0.25\sigma$. With the measured slope $d\ln(hr_d)/d\ln h = 0.49$ (`fixround_diagnostics.json`), a gate-admitted offset propagates to $|\Delta H_0|$ up to 0.25 (DR2) and $0.45 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DR1), larger than the strict window of 0.15 in both cases and, for DR1, larger than the soft window of 0.33 . The T2 PARTIAL is *consistent with* this mechanism, but the numbers do not establish it: the G2 offset from DR1’s Sec. 8 value $hr_d = 101.8$ is 0.134 Mpc (pull 0.10), which implies $0.183 \text{ km s}^{-1} \text{ Mpc}^{-1}$; from DR1’s Sec. 6 value 101.9 it implies $0.046 \text{ km s}^{-1} \text{ Mpc}^{-1}$; across the $\pm 0.05 \text{ Mpc}$ rounding interval of 101.8 alone it implies $0.11\text{--}0.25 \text{ km s}^{-1} \text{ Mpc}^{-1}$; and the 2000-step G2 chain (ESS 2115) carries a Monte Carlo error of 0.028 Mpc on its mean, i.e. $\pm 0.038 \text{ km s}^{-1} \text{ Mpc}^{-1}$ on the implied value (`round2_response_checks.json`, key `g2_attribution`). The mechanism can therefore account for anywhere from about 0.05 to $0.25 \text{ km s}^{-1} \text{ Mpc}^{-1}$ of the observed 0.167 , and the closeness of 0.183 to 0.167 is not evidence for it. The T1 strict PASS is likewise only *consistent with* G1 landing at -0.020 Mpc (implied $-0.027 \text{ km s}^{-1} \text{ Mpc}^{-1}$); we do not attribute it to that offset. PASS versus PARTIAL therefore does not isolate the BBN/ r_d step. No tolerance was changed after the fits; the verdicts are those the preregistered rule gives for the recorded run (computed by `results/cosmo_synthesis/round1_response_checks.py`), and Sec. 4.3 shows that the primary T2 label is not stable under re-execution. An independent DESI DR2 BAO+BBN analysis with a full BBN likelihood [33] finds $h = 0.6833^{+0.0048}_{-0.0053}$ with the PRIMAT network and “a modest spread in h across the three reaction networks considered here, up to roughly 0.5σ ”, noting that “the DESI DR2 analysis uses a prior on the baryon density from just one reaction network”. A reaction-network systematic of that size is larger than our strict window (0.26σ); it plays no role in our verdicts, which reproduce DESI’s analysis with DESI’s prior, but a systematic of that size is comparable to or larger than the distance between the PASS/PARTIAL boundary and our result (the two σ units differ: theirs is their own $\sigma(h)$, ours DESI’s).

The DESI DR2 and eBOSS preregistrations were not amended after fitting. The eBOSS preregistration carries an amendment field (before any fit) whose recorded time is later than the file’s mtime (Sec. 6). None of the preregistration files was committed to git before its fit, so for all of them the file mtime is the only evidence of ordering.

3.3 Controls, including a subtle-bug control

DESI DR2: 5 preregistered positive controls (PC0–PC4; PC0 has quadrature and Gauss–Legendre entries, so `controls.json` has 6 keys) and 5 negative controls. The noisy-mock control used 500

draws, with an Ω_m pull mean of -0.002 and standard deviation of 0.970 . The wrong-model (EdS) control gave $\chi^2 = 1420.2$. The scrambled-covariance control passed its preregistered median-PTE rule (median 3.4×10^{-6}), but 6 of 20 individual scrambles gave $\text{PTE} > 10^{-3}$, so it tests covariance ordering only in aggregate. **H0**: noiseless mock (P1, $|\Delta H_0| = 0.0041$), 100 noisy mocks (P2, H_0 pull mean -0.064 , s.d. 1.023 ; report-only, because the preregistered rule covers H_0 only: Ω_m pull mean -0.131 with standard error 0.094 , s.d. 0.944), gates G1/G2 (DESI BAO-only reproductions), predictor agreement (P4, 1.7×10^{-6}), scrambled data (N1), EdS (N2, $\chi^2 = 1420.2$) and no-BBN (N3, $\sigma(H_0)$ grows by a factor of 21.0). **eBOSS**: a grid-likelihood instrument check, an injection test over 200 realizations on the SDSS Gaussian-summary variant and DESI, and negative controls. An injected $\times 0.93$ rescaling of all ratios gives $N_\sigma = 12.1$.

Scrambled data and wrong models fail in any pipeline, so they do not show that a pipeline is right. After review, the H0 run added **N4**, a deliberately subtle convention error: ω_ν not subtracted from ω_{cdm} in the CAMB r_d . Its status is recorded as “post-hoc (referee fix round), not preregistered, not in the verdict’s controls condition”. It moves the MAP H_0 by $0.164 \text{ km s}^{-1} \text{ Mpc}^{-1}$ while changing χ^2 by only -5.2×10^{-5} . The data cannot see this error (the wrong model even has the lower χ^2). Only the external target can, and only marginally: the strict window of $0.15 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is exceeded by $0.014 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.02 published σ), which is 1.3 times the T1 posterior-mean Monte Carlo error of $0.0107 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The shift is measured between MAP points, whereas the verdicts use posterior means. We think controls of this kind, which the likelihood cannot detect, are the important ones for agent-written analyses.

3.4 Lean 4 formalization and the dual-environment check

Each run states the algebraic skeleton of its model in Lean 4: positivity and monotonicity of $E(z)$, the distance-duality identity, positivity and monotonicity of the Aubourg r_d formula, a $(h, r_d) \rightarrow hr_d$ degeneracy identity, identifiability of h from $hr_d(h)$ for the fitting-formula r_d , and symmetry, non-negativity and zero-iff properties of the tension statistic. These theorems are elementary; 7 of the 28 (`rd_numerator_pos`, `DH_over_rd_eq_hrd`, `DH_over_rd_degenerate`, `gate_reparam_exact`, `tension1D_symm`, `tension1D_nonneg`, `surveyTension_symm`) are short identities or positivity statements (“short” refers to their mathematical content; some proofs take three tactics), none vacuous. They check that stated properties of the formulas used hold. **They do not verify the fits, the data handling or the numerics.** The statements appear verbatim in Appendix A; the docstrings, which carry the scope limits, are stripped there, so we state the limits here. For the H_0 run: the degeneracy theorems are proved for D_H/r_d only and at fixed $E(z)$, whereas in the BBN fits E depends on h through radiation; the identifiability theorem (`hrd_strictMonoOn_h`, via `hrd_mono_generic`) covers only the *secondary* Aubourg r_d , on the set $\omega_\nu \leq (1-2a)\Omega_m h^2$, which excludes part of the sampled secondary prior (stated in the Lean file’s header and measured in `fixround_diagnostics.json`, key `omega_m_threshold`); nothing is proved about the CAMB r_d behind the headline $H_0 = 68.545$ PASS. For DR2, the preregistration planned the domain $z > -1$; the shipped statements use $z \geq 0$ (Sec. 3.6). In revision we tested the preregistered domain for the two positivity theorems: `E2_pos` and `E_pos` restated with hypothesis $-1 < z$, with the `E2_pos` tactic script byte-identical and the `E_pos` script differing only in the name of the lemma it calls, compile in the pinned build (rc 0) with whitelist-only axioms (`results/cosmo_synthesis/dr2_zgtm1_check.lean`, sha256 prefix `ca31d646`). For these two theorems the preregistered domain is recoverable at no proof cost; the shipped file keeps $z \geq 0$ only so that the model audit stays bound to the audited bytes. The monotonicity and D_C theorems were not tested on $z > -1$ (the preregistration fixed no domain for them; its $z > -1$ applies to the positivity statement only). In `BAO_FlatLCDM.lean`, the hypothesis $z \neq -1$ of `distance_duality` is inert in Lean (both sides are 0 at $z = -1$ because $x/0 = 0$), although its

docstring suggests it is needed.

The binding project rule routes Lean through `anse/formal/lean_runner.py`, which could not gate a new file from a worktree. Every run therefore used `lake env lean <file>` plus in-file `#print axioms`, which is a documented deviation. After the runs, all four files were re-checked in two environments (Table 7); both report Lean 4.34.0 – rc2 (toolchain `leanprover/lean4:v4.34.0-rc2`) and Mathlib v4.34.0-rc2 at commit 85e3a25e006c, read from each environment’s `lake-manifest.json`. Environment 1 is the project’s pinned, partially built Mathlib (3431 Mathlib `.olean` files for 8370 sources). Environment 2 is the LeanMaster project’s fully built copy of the same Mathlib commit (8370 `.olean` files). The two environments therefore share the Lean binary and the Mathlib version; the second check tests independence from the partial build, not from the compiler or the library. In environment 1, 28 of 28 axiom outputs list only `{propext, Classical.choice, Quot.sound}`. In environment 2, 22 were checked and clean. The remaining 6 (`DESI_DR2_wCDM.lean`) were not checked, because that file imports `ANSE.BAO_FlatLCDM`, which environment 2 does not provide. Lean stopped with “unknown module prefix ‘ANSE’” before elaborating any declaration. This is `BLOCKED_ENV`, not a proof failure. Accepted file×environment pairs: 7 of 8. **Negative controls:** (1) a copy of `BAO_FlatLCDM.lean` with the proof of `E_pos` replaced by `sorry` exited with rc 0 / 0, while `#print axioms` showed `sorryAx` (yes). (2) A copy that declares axiom `smuggled_E_pos_fact : False` and proves `E_pos` from it exited with rc 0 / 0; the smuggled axiom appeared in the `#print axioms` output (yes) and the whitelist rejected it (yes). An exit code alone would have accepted both. **Correction of a hand-edited record.** The first version of the checker counted the substring `#print axioms` including a comment line, rejected `BAO_FlatLCDM` in both environments, and the output JSON was then corrected by hand (keys added without a re-run; kept as `lean_dual_check_v1_handedited.json`, sha256 prefix `bc09047fb601`). The round-1 formal referee found this. The checker now counts `#print axioms` commands on comment-stripped source, computes every key the paper reads, adds the axiom-smuggling control and records the toolchain; all numbers in this section and Table 7 come from that re-run. All kernel checks are single-file `lake env lean` runs. The three new modules are not yet part of the project’s `lake build`: the worktree’s `formal/ANSE.lean` imports all four, but no build of that root is recorded, and the main checkout’s `formal/ANSE.lean` imports `ANSE.BAO_FlatLCDM` 2 times and none of the three new modules (0 imports). Wall times are single measurements on a shared machine (load average at the end of the run: 4.34 / 4.42 / 4.84) and are not benchmarks.

3.5 Elenchus tier-capped ledger

Claims are filed with tier $X < C < L < B < A$. A is kernel-checked Lean, L is a literature citation and C is a conjecture. **Our Tier B is not Elenchus’s Tier B.** In the cited tool, B is an identity verified in exact rational ($\mathbb{Q}/\mathbb{Z}$) arithmetic, and floats, sampling and model output are filed at X, which “may never support a claim” [8]. All 34 Tier B rows in our four ledgers (1/8/17/8) are seeded floating-point or Monte Carlo harness outputs on sha256-verified inputs (emcee posterior moments, CAMB/CLASS backgrounds, control χ^2). Under the tool’s own semantics they would be X, and by dependency closure so would the computed L comparison rows that rest on them. The gate cannot see this because `kind` is self-declared. The H0 and eBOSS ledger READMEs disclose the redefinition; the DR1 and DR2 ledger READMEs do not. No number changes; only the label “B” is borrowed from a ladder that would not grant it, and the round-1/2 closure mutation tests (tier inversion, refiling a B row at C) presuppose our redefinition. A claim’s tier is capped by the tiers of its dependencies and by its evidence kind, and each evidence blob is addressed by its sha256. The gate (`ledger.py`) returns rc 1 on any block or flag. On a Tier A row, “flag” means the kernel

checked the proof but no one has certified that the statement means what its gloss says. **The gate only type-checks the audit field**, so a model-written audit turns rc 1 into rc 0. In every run the audits are the model referee’s and are labelled as such, so rc 0 here means *model-audited*, not human-audited (Table 8). The DR1 ledger keeps its three Tier A rows unaudited (rc 1). The H0 ledger has rc 1 because two theorems added in the fix round were never reviewed. Mutation controls were run against the three preregistered runs’ ledgers: a tier inversion, a tampered blob and a kind overclaim. The gate caught each one. **What the gate does not check.** It binds each evidence digest to bytes, not a claim to its evidence: the round-1 formal referee pointed a Tier A row’s digest at a different, intact blob in the same evidence directory, and the gate returned rc 0/0 (DR2/eBOSS); an empty audit object {} also cleared the flag (rc 0/0). The gate also never parses the content of a `lean_axioms` blob: in a mutation control devised by the round-3 formal referee and re-run here on copies (`results/cosmo_synthesis/round3_m6_sorry_blob_mutation.py`), rewriting one Tier A row’s own blob so that its axiom footprint lists `sorryAx`, re-hashing and repointing the row left the gate’s rc unchanged in 3 of 3 ledgers. The kernel-checked meaning of Tier A therefore rests on whoever wrote the blob, not on the gate. Claim-to-blob correspondence is checked only by reading; the round-3 formal referee did that reading programmatically and found all 28 Tier A blobs (one per theorem) content-consistent with the shipped Lean files. The three ledgers use three audit schemas (`auditor_kind`, `auditor_is_person`, free-text by); Table 8 therefore counts an audit only if it is a non-empty object naming an auditor, and a further column counts audits whose recorded source hash equals the current Lean file. For DR2 the audited bytes are the current file. For eBOSS the audit is bound to an earlier hash, and the ledger records the current hash and the change (a docstring). For H_0 , the 8 audits are bound to source hash `c0663994...`, not the current `2b653083...`, and the ledger names neither the current hash nor the change, against the project rule that an audit is bound to the Lean file’s hash. The audited bytes were never committed. In the only surviving pre-fix copy (a `sorry` scratch file, now kept as `results/cosmo_synthesis/h0_prefix_scratch_copy.lean`, not hash-linked to the audited bytes), 8 of the 8 audited statements are textually identical to the current ones (`round1_response_checks.json`, key `h0_audit_binding`). The ledger lies outside this synthesis’ write scope and was not changed. Some computed comparison rows (e.g. DR2-L-0006, BBNH0-L-0008, EVD-L-0006) are filed with kind `citation`; the tier cap is unaffected because it also follows from dependency closure, but the kind label does not describe how the claim was established.

3.6 Adversarial referee and fix round

A model referee re-ran each preregistered analysis and read the Lean statements. DR2: verdict minor, 8 issues. H0: verdict minor, 8 issues. eBOSS: 7 issues in round 1 (8 of 9 fix items fixed; the fix list has more items than the report because issue 7 was split into 7a/7b/7c) and 9 in round 2 (7 of 10 fixed). The eBOSS ledger’s rc 0 (Table 8) comes from fix item `r2_8b`, which recorded the round-2 referee’s statement audit on the Tier A rows “per the orchestrator’s fix-round instruction”. The round-2 referee had described the reverted rc 1 state as “the honest outcome”; rc 0 was therefore reached over the referee’s stated preference. The referees found real problems:

- a citation overreach (DR1 does not apply its parameter-difference statistic to SDSS);
- a vacuous negative control (the SDSS-baseline EdS test passed because the likelihood returned $+\infty$);
- a combination guard that was only a filename substring match;

- a DR2 Lean statement that had drifted from the preregistered $z > -1$ to $z \geq 0$ (left in the shipped file so that the audit stays bound to the audited bytes; in revision, $z > -1$ was tested for `E2_pos` and `E_pos` and proves with the same tactics, Sec. 3.4);
- a preregistered H_0 monotonicity theorem that had been replaced by easier identities;
- unverifiable model attributions in bylines.

In the DR2 fix round, the chains were regenerated and compared bit for bit with the originals (identical: yes).

3.7 Cross-language cross-check in Rust/CVODE

The distance integrals and the DR2 flat- Λ CDM χ^2 fit were ported to the `qf-bao-distances` crate of rusty-SUNDIALS (commit `c46a680`; PR #61, recorded by `gh pr view` as merged at 2026-09-27T17:29:02Z with merge commit `7c9c3c3`), on that repository’s own Rust reimplementations of CVODE (not the LLNL library). The CVODE path gives $\Omega_m = 0.29743 \pm 0.00861$ and $hr_d = 101.543 \pm 0.735$ Mpc with $\chi^2_{\min} = 10.2710$. The quadrature path differs from our Python χ^2 minimum by -1.8×10^{-6} in Ω_m , which is at the README’s printed precision. The CVODE path differs by -3.2×10^{-5} . That is a real offset between the CVODE and quadrature integrators, larger than the rounding, but within the crate’s own test bound of 1% of σ for CVODE versus quadrature. This comparison is between χ^2 minima with Fisher errors, not the emcee posterior means of Table 2. The crate’s tests report 12 passing and 1 deliberately ignored.

The port exposed defects in this Rust CVODE reimplementations. The README (last changed in `c46a680`) records these measurements:

- the Einstein–de Sitter closed-form bound of 10^{-7} is not met by CVODE BDF at `rtol` 10^{-7} (measured 7.4×10^{-7});
- many outputs from one BDF run lose accuracy;
- `rtol` $\leq 10^{-8}$ fails;
- `Method::Adams` never leaves order 1, because its coefficient routine is a placeholder (error 2.1×10^{-4} at `rtol` 10^{-7}).

The later commit `d22cf01` (PR #60) identifies a cause: output-time truncation shifted the Nordsieck history instead of rescaling it. That commit reports a one-line fix and says `rtol` $\leq 10^{-8}$ now works. The README measurements predate that fix and were not repeated afterwards. The Adams order-1 defect is not addressed by `d22cf01` and remains open.

4 Results

Pulls are $(\text{ours} - \text{published}) / \sigma_{\text{published}}$. Our fits use the same data, priors and estimator as the published analyses, so a pull is a measure of reproduction accuracy, not an independent-sample significance (the DR2 Monte Carlo error on the Ω_m mean is 0.007σ). Figure 1 summarizes all targets. Each point’s grey band is its own preregistered tolerance, not an error bar: 0.5σ for DR2 and eBOSS; 0.25σ for the H_0 Ω_m targets and gates; and, for H_0 , the absolute strict window of $0.15 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which is 0.26σ (T1) or 0.19σ (T2).

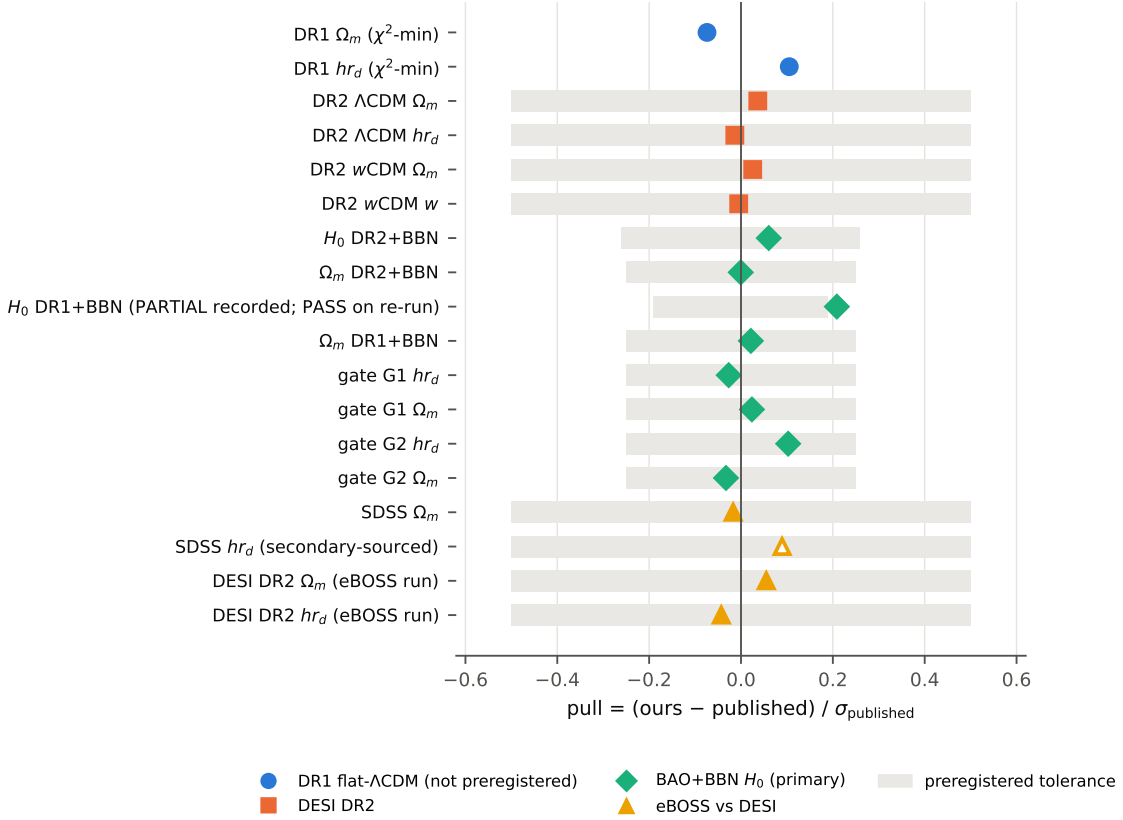


Figure 1: Pulls of every numeric target, from the result JSONs by `make_figures.py`. Grey bands show each target’s preregistered tolerance; they are tolerances, not error bars, and the pulls are not independent-sample significances (same data and priors on both sides). The DR1 run was not preregistered and has none. The hollow triangle (SDSS hr_d) marks a target taken from a secondary source, which is excluded from the eBOSS tolerance rule. H_0 secondary (fitting-formula) pulls are omitted here. They are listed in Table 3 and shown in Fig. 2.

4.1 DESI DR1, flat Λ CDM

Table 1. Two independently coded predictors agree exactly, and a brute-force 141×141 grid (steps 0.00143 in Ω_m and 0.179 Mpc in hr_d) gives $\Omega_m = 0.2957$ and $hr_d = 101.79$ ($\chi^2 = 12.7562$), 1.3 and 0.8 grid steps from the optimizer minimum along the degeneracy, i.e. within about one grid step. Table 1 compares our χ^2 minimum with DESI’s posterior mean (eq. 4.1). The fetched DR1 paper also gives the *matched* target, its best-fit model: $\Omega_m = 0.294$ and $H_0 r_d = 101.94 \times 100 \text{ km s}^{-1}$ (Fig. 1 caption), with $\chi^2 = 12.66$ for 10 degrees of freedom (Sec. 3.2) [1]. Our $\Omega_m = 0.2939$ and $hr_d = 101.94$ agree with it to the printed precision. Our $\chi^2_{\min} = 12.74$ for 10 degrees of freedom is higher by 0.081 on the same public data vector and covariance (sha256 in Table 9). We tested background physics as the cause (`results/cosmo_synthesis/dr1_chi2_offset_check.py`): adding radiation, or replacing the analytic distances by a CAMB background with massless neutrinos or with one 0.06 eV neutrino, changes $\chi^2_{\min}$ by at most 0.003, leaving an offset of 0.078 in the DESI-like CAMB case. **The offset is unexplained**; it may come from a difference in the likelihood inputs or the minimizer used by DESI, which we cannot inspect. The DR2 run later re-fitted DR1 with the posterior-mean estimator and obtained $\Omega_m = 0.29498$ (pull -0.001) and $hr_d = 101.89$

(pull 0.067).

Table 1: DESI DR1 flat Λ CDM. Source: DESI DR1 2024 VI, arXiv:2404.03002, quoted from fetched paper text (eq. 4.1).

Quantity	Ours (χ^2 -min)	σ	Target	σ_{target}	Pull
Ω_m	0.2939	–	0.295	0.015	−0.074
hr_d [Mpc]	101.94	–	101.8	1.3	0.105

4.2 DESI DR2, Λ CDM and w CDM

Table 2. Primary estimator: emcee posterior mean and standard deviation (20000 steps per chain, converged under the 50τ rule). The largest emcee-minus-grid difference is 0.009 emcee σ . The correlation $r(\Omega_m, hr_d) = -0.923$ compares with the published -0.92 , and the best-fit χ^2 is 10.27 for 11 degrees of freedom. The w CDM $hr_d = 99.83 \pm 1.73$ Mpc is report-only, because no published value was found.

Table 2: DESI DR2 BAO only (`results/desi_dr2_bao/fit.json`).

Quantity	Ours	σ	Target	σ_t	Pull	Source
Λ CDM Ω_m	0.29781	0.00856	0.2975	0.0086	0.037	arXiv:2503.14738v3 eq.(17) p.19 and Table V (DESI, LCDM); identical in v1 eq.(17)
Λ CDM hr_d [Mpc]	101.530	0.732	101.54	0.73	−0.013	arXiv:2503.14738v3 eq.(17) p.19; identical in v1
w CDM Ω_m	0.29713	0.00896	0.2969	0.0089	0.026	arXiv:2503.14738v3 Table V, w CDM row ‘DESI’
w CDM w	−0.9164	0.0787	−0.916	0.078	−0.005	arXiv:2503.14738v3 Table V, w CDM row ‘DESI’

4.3 BAO+BBN inverse distance ladder

Table 3; sources in Table 4. The primary DR2 result passes every PASS condition. The DR1 check is PARTIAL: $|\Delta H_0| = 0.167$ exceeds the strict window by $0.017 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which is comparable to our posterior-mean Monte Carlo error of 0.014. DESI’s own posterior-mean sampling error is not published; its stated convergence floor ($\text{ESS} \gtrsim 10^3$, [1, Sec. 2.5]) implies up to about 0.025 on the quoted 68.53, plus ± 0.005 rounding. The combined resolution is then about 0.029, so the margin is 0.58 of it and the strict/soft boundary is not statistically resolved (as the `T2_boundary_resolution` note in `fit.json` already states). The preregistered rule gives PARTIAL for the recorded run, and PARTIAL is what that run reports; re-execution under a different numerical environment gives PASS (next paragraph), so we do not present PARTIAL as a stable property of the analysis. A post-hoc diagnostic, which does *not* change the verdict: DR1 quotes $hr_d = 101.8$ in one section and 101.9 in another. Propagating these gives implied H_0 excesses of 0.183 and $0.046 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.11–0.25 across the rounding of the first, ± 0.038 from the G2 chain). The G2 gate used the first value. As shown in Sec. 3.2, the gate tolerance itself admits offsets of this size. *Code deviation:* `scripts/bao_bbn_h0/fit_real.py` (line 262) passes gate G1 into `verdict_T2`, although the pre-registration defines gate G2 for the DR1 check. G2 passed (its `passed` field in `fit.json`), so the T2 verdict is unchanged; the script is frozen and outside this synthesis’ write scope.

Numerical-environment dependence (found by the round-3 referee, confirmed here). The machine has two Python environments: `venv-pta` (Python 3.10.12, numpy 1.26.4, scipy 1.15.3) and `venv-cosmo` (Python 3.11.16, numpy 2.4.6, scipy 1.17.1), both with emcee 3.1.6. No run JSON records its interpreter. We re-ran the frozen `fit_real.py` in isolated copies under both, changing only the chain output directory (`results/cosmo_synthesis/round3_env_rerun_setup.py`, `round3_response_checks.py`; the two re-run outputs are kept as `round3_rerun_fit_venv_{pta,cosmo}.json`, hashes in Table 10). Under `venv-pta`, 0 of 943 numeric leaves of `fit.json` differ: the recorded H_0 run was produced by `venv-pta`. Under `venv-cosmo`, 751 of 943 differ. MAP H_0 values agree to 9.2×10^{-8} , but the posterior means move by up to 1.5 Monte Carlo errors: primary T1 $H_0 = 68.549$ (PASS), **primary T2** $H_0 = 68.675$, $|\Delta H_0| = 0.1455$, i.e. **PASS** instead of the recorded 0.167 (PARTIAL), secondary T1 PASS and secondary T2 $|\Delta H_0| = 0.1725$ (PARTIAL). Gate G2’s hr_d mean becomes 101.90 Mpc; the implied H_0 shift, computed as in Sec. 3.2, becomes $0.140 \text{ km s}^{-1} \text{ Mpc}^{-1}$, inside the range quoted there. Integer leaves (chain lengths and burn-in) also differ, because the chains stop adaptively. The round-3 referee found the DR2 Λ CDM chain bit-identical under both interpreters; we did not repeat that check. The preregistration names no interpreter, so both executions are equally valid applications of the preregistered rule. The primary T2 label is therefore **PARTIAL in the recorded run and PASS on re-execution**; its margin over the strict window (0.017) is about one Monte Carlo error, and chains of the recorded length cannot resolve it. A preregistered verdict whose margin can be one MC error should either fix a chain long enough that the MC error is much smaller than the margin, or be stated on a deterministic estimator (the MAP, or a posterior mean with a preregistered MC floor).

Formula-versus-CAMB shift. Exact CAMB versus the fitting formula moves the MAP H_0 by -0.0419 (DR2) and -0.0411 (DR1), stable to 10^{-7} across both environments. The posterior-mean differences, -0.034 ± 0.015 (DR2) and -0.0044 ± 0.020 (DR1) in the recorded run and -0.029 / -0.027 under `venv-cosmo`, are dominated by Monte Carlo noise; the smaller DR1 value is not a smaller systematic. The largest fitting-formula error measured against CAMB is 0.029 %. Aubourg et al. state 0.021% accuracy for $N_{\text{eff}} = 3.046$ and parameters within 3σ of Planck [5]; 2 of our 5 validation points exceed that figure, including the largest one at $(\omega_b, \omega_{\text{cdm}}) = (0.02218, 0.119)$, which lies inside the proxy validity window used in the preregistration. Our CAMB table uses $N_{\text{eff}} = 3.044$, whereas Aubourg’s statement assumes 3.046. Re-running CAMB (v2.0.4) at $N_{\text{eff}} = 3.046$ for the same points (`round2_response_checks.json`, key `aubourg_neff`) shows that r_d at 3.044 exceeds r_d at 3.046 by $(6.7\text{--}6.9) \times 10^{-5}$, close to the 6.5×10^{-5} of Aubourg’s eq. (17) factor. Against the 3.046 CAMB values, 1 of the 5 points still exceeds 0.021% (largest 0.0224%). Part of the remainder is a convention in the validation code (found by the round-3 referee): `rd_camb_table.py` compares CAMB and eq. (16) at equal $\Omega_m h^2$, which is what the fit samples, and so evaluates eq. (16) at $\omega_{cb} = \omega_b + \omega_{\text{cdm}} + (\omega_{\nu}^{\text{CAMB}} - \omega_{\nu}^{\text{Aubourg}})$, i.e. 2.9×10^{-6} above the CAMB grid point’s $\omega_b + \omega_{\text{cdm}}$. Aubourg’s accuracy statement is at given ω_{cb} . Evaluated at the grid point’s own $\omega_b + \omega_{\text{cdm}}$, the largest deviation from the $N_{\text{eff}} = 3.046$ CAMB values is 0.0219% (1 point above 0.021%; `round3_response_checks.json`, key `aubourg_omega_cb`). The N_{eff} convention and this ω_{cb} convention together explain most of the excess; the residual of order 10^{-5} was not traced further. The preregistered comparison (against the 3.044 table) is the one reported above. The CAMB anchor at the Planck 2018 best fit gives $r_{\text{drag}} = 147.049$ Mpc; the preregistration cites 147.049.

4.4 SDSS/eBOSS versus DESI DR2

Table 5. The SDSS (Ω_m, hr_d) correlation we find is -0.880 ; SDSS does not publish it. The baseline and Gaussian-summary SDSS likelihoods differ by 0.032σ in Ω_m . The DESI numbers here come

Table 3: BAO+BBN H_0 (`results/bao_bbn_h0/fit.json`, key `pulls`; recorded run, `venv-pta`). Values in $\text{km s}^{-1} \text{Mpc}^{-1}$ for H_0 and Mpc for hr_d . “secondary” is the preregistered fitting-formula analysis and is identical to the hash-locked first attempt in the same numerical environment. The “shared” gate rows are BAO-only reproductions. [‡]Environment-dependent: re-executing the frozen script under `venv-cosmo` gives $|\Delta H_0| = 0.1455$ (strict rule met); the boundary is within Monte Carlo resolution (Sec. 4.3).

Analysis	Target	Ours	σ	Target	σ_t	Pull	Strict rule met
primary	T1_H0	68.545	0.594	68.51	0.58	0.060	yes
primary	T1b_Omega_m	0.2977	0.0085	0.2977	0.0086	−0.001	yes
primary	T2_H0	68.697	0.809	68.53	0.80	0.209	no (soft: yes) [re-run: yes] [‡]
primary	T2b_Omega_m	0.2953	0.0146	0.295	0.015	0.022	yes
secondary	T1_H0	68.579	0.601	68.51	0.58	0.119	yes
secondary	T1b_Omega_m	0.2976	0.0086	0.2977	0.0086	−0.015	yes
secondary	T2_H0	68.701	0.815	68.53	0.80	0.214	no (soft: yes)
secondary	T2b_Omega_m	0.2946	0.0143	0.295	0.015	−0.028	yes
shared	G1_hrd	101.52	0.74	101.54	0.73	−0.027	yes
shared	G1_Omega_m	0.2977	0.0087	0.2975	0.0086	0.024	yes
shared	G2_hrd	101.93	1.27	101.80	1.30	0.103	yes
shared	G2_Omega_m	0.2945	0.0145	0.295	0.015	−0.033	yes

from an independent re-fit with shorter chains, so they differ slightly from Table 2.

4.5 Derived quantities not quoted in the fetched DESI text

Table 6. DR2 does test consistency with DR1 and with SDSS, but at the level of distances. It compares DR1 and DR2 bin by bin with a correlated joint uncertainty and a KS test, and compares SDSS with DESI per redshift bin in α_{iso} and α_{AP} [2, Sec. III.C]. In the fetched text we found no parameter-level (Ω_m, hr_d) number for either comparison. The quantities below complement those checks and do not replace them.

(a) DR1 and DR2 share data. We use the nested uncertainty on their difference, $\sigma_{\text{nested}}^2 = \sigma_{\text{DR1}}^2 - \sigma_{\text{DR2}}^2$, which gives $0.237\sigma_{\text{nested}}$. DR2 describes its own DR1–DR2 comparison as assuming perfect correlation, but computes $\sigma^2 = \sigma_{\text{DR2}}^2 + \sigma_{\text{DR1}}^2 - 2C\sigma_{\text{DR1}}\sigma_{\text{DR2}}$ with $C = \sqrt{N_{\text{DR1}}/N_{\text{DR2}}}$ per galaxy/quasar tracer ($C = 0.61$ for $\text{Ly}\alpha$) [2, Sec. III.C.1 and footnote 12]; with $\sigma \propto N^{-1/2}$ this is the nested form for the galaxy and quasar bins, which is all the parameter-level argument needs. Taking $C = 1$ literally at the parameter level gives $\sigma = |\sigma_{\text{DR1}} - \sigma_{\text{DR2}}| = 0.0061$ and 0.46σ , and the independent convention gives 0.167σ . The $\sigma \propto N^{-1/2}$ step is only approximate: Kim, Mota & Tamosiunas [34], who model the DR1-inside-DR2 nesting explicitly with a year-scaling decomposition (their Appendix B.2), report that the published per-bin covariances “straddle” the ratio $C_{\text{DR2}} = C_{\text{DR1}}/3$ that pure year scaling implies, at 0.22–0.49 (median 0.32). Our σ_{nested} is therefore an approximation, bracketed by the $C = 1$ and independent conventions. All three conventions agree that the shift is insignificant. The published means imply $\Delta\Omega_m = 0.0025$.

(b) The SDSS–DESI DR2 tension applies DESI’s parameter-difference statistic (eq. 18 of [2]) to BAO-only (Ω_m, hr_d) under the independent-survey assumption. DR2 states that the two surveys are consistent but gives no number for this pair of parameters. For DR1, the DESI paper gives a qualitative parameter-level statement: “no significant difference in Ω_m and a shift of just $\sim 1\sigma$ in $r_d h$ ” [1, Sec. 3.3]. Two other papers compare SDSS and DESI and reach different-sounding conclusions because they ask different questions. Ghosh & Bengaly [24] reconstruct $H(z)$ and $q(z)$ non-parametrically (Gaussian processes) from D_H/r_d alone, for DESI DR1, with a Planck r_d , and

Table 4: H0 target sources, read from `preregistration.json` (not editable after the fit). For G2 the string names the DR1 abstract, but the fetched abstract quotes only Ω_m and H_0 ; $r_d h = 101.8 \pm 1.3$ is in eq. (4.1) and Sec. 8 of [1].

Target	Source (preregistration)
T1	arXiv:2503.14738 eq. (19) and Table V row DESI+BBN (no CMB theta*)
T1b	arXiv:2503.14738 Table V
T2	arXiv:2404.03002 eq. (4.4), Table 3 (no CMB theta*)
T2b	arXiv:2404.03002 Table 3
G1	arXiv:2503.14738 eq. (17)
G2	arXiv:2404.03002 abstract/Sec. 8

Table 5: eBOSS vs DESI (`results/eboss_vs_desi/fit.json`). [†]Taken from a secondary source (DR1 quoting SDSS). We did not find it in the fetched pages of the primary paper, so it is not counted toward the tolerance rule.

Quantity	Ours	σ	Target	σ_t	Pull	Source
SDSS Ω_m	0.2987	0.0164	0.299	0.016	-0.017	arXiv:2007.08991 Table IV (BAO, flat Λ CDM): $\Omega_{DE} = 0.701^{+0.017}_{-0.015} \rightarrow \Omega_m = 1 - \Omega_{DE} = 0.299^{+0.015}_{-0.017}$; symmetric 0.016 also quoted by arXiv:2404.03002 Sec. 4.1
SDSS hr_d [Mpc] [†]	100.52	1.29	100.40	1.30	0.090	arXiv:2404.03002 Sec. 4.1 citing [139]=arXiv:2007.08991
DESI DR2 Ω_m	0.2980	0.0087	0.2975	0.0086	0.055	arXiv:2503.14738 eq. (17)
DESI DR2 hr_d [Mpc]	101.51	0.75	101.54	0.73	-0.043	arXiv:2503.14738 eq. (17); corr r = -0.92

find the reconstructions “significantly inconsistent”; this is not a parameter-level test in (Ω_m, hr_d) . Ferri, Ruchika & Melchiorri [25] compare SDSS DR16 and DESI DR2 combined with Planck in the CPL model and find offsets in w_0 and q_0 of about 1.1σ . Our number is narrower: BAO alone, flat Λ CDM, (Ω_m, hr_d) , eq. (18). Before any fit, the preregistration had bounded N_σ to 0.32–1.56 from published means alone, so the value is an application of a known statistic, not a surprise. The 1D differences are 0.041σ (Ω_m) and 0.668σ (hr_d). Because the surveys overlap, the independent-survey N_σ is a lower bound only under the assumption that the cross-covariance is positive; with $X + X^T$, $X = \rho\sqrt{C_S}\sqrt{C_D}$, subtracted from $C_S + C_D$ (the covariance of a difference), N_σ rises to 1.34 at $\rho = 0.57$ (PTE 0.18). The value 0.57 is DR2’s estimate for one bin pair (LRG2 and eBOSS LRG), the only pair for which DR2 quotes C [2, Sec. III.C.2]; applying it to the whole parameter-level covariance is an illustration, not an estimate. DR2 uses the same lower-bound logic for its own distance-level comparison: for that bin pair the discrepancy is $\sim 2.6\sigma$ with $C \approx 0.57$, and “although the assumption of no correlation is less plausible, it sets a lower limit of the discrepancy at the 1.9σ level” [2, Sec. III.C.2] (2.3σ and 1.5σ with DESI’s reanalysis of SDSS; our eBOSS run uses the published SDSS DR16 likelihoods, to which the 1.9 – 2.6σ figures refer). Our parameter-level $N_\sigma = 0.877$ therefore does not mean that there is no tension anywhere: the LRG2/eBOSS-LRG bin pair (the one DR1 had flagged at $\sim 3\sigma$) is at 1.9 – 2.6σ by DR2’s own numbers, and the two-parameter

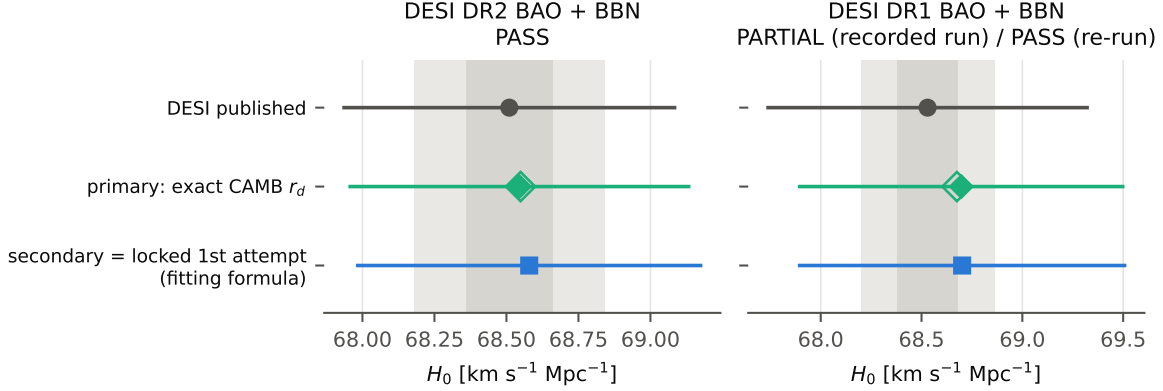


Figure 2: H_0 from BAO+BBN. The primary analysis uses the exact CAMB r_d (amendment 1). The secondary analysis uses the preregistered fitting formula and, in the recorded environment, equals the hash-locked, unread first attempt. Filled symbols: recorded run (`venv-pt`); hollow diamond: the primary posterior mean when the same frozen script is re-executed under `venv-cosmo` (Sec. 4.3). The dark band is the strict window (± 0.15) and the light band the soft window (± 0.33) around the DESI value.

BAO-only fit averages over bins.

(c) The CAMB-versus-formula H_0 shift, measured deterministically between MAP points as -0.0419 (DR2) and $-0.0411 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DR1), equals the propagation of the *measured* formula error near the posterior (-0.029% at the first validation point, which gives -0.0408 and -0.0408 through $\Delta \ln h = -\epsilon/(d \ln h r_d / d \ln h)$ with the slopes of `fixround_diagnostics.json`). Aubourg et al.’s stated 0.021% would give only about -0.029 . The shift therefore does not confirm the stated accuracy figure; it is consistent with the exceedances of 0.021% reported in Sec. 4.3, and it is not a new quantity.

5 What is and is not new

Not new. Every cosmological parameter here reproduces a published value from the same public likelihood files, using methods the DESI and SDSS papers describe. Nothing here is a new measurement or a new constraint.

Prior work we found. The first draft’s search used four alphaXiv queries (LLM agents reproducing cosmology analyses; formal verification of physics in Lean or other proof assistants; HepLean/PhysLean; benchmarks of agents reproducing astrophysics papers); their exact strings and dates were not recorded. For this revision we ran two further queries, on SDSS–DESI consistency and on blind analysis and preregistration for agents, and recorded them exactly (`literature_search_log_round1.json` in `results/cosmo_synthesis/`). Those queries missed two axes, pre-run cryptographic commitment of cosmological analyses and autonomous analysis agents with model review and an unblinding gate. The round-2 novelty referee supplied the papers below by arXiv id; we fetched and read each before citing it (`literature_search_log_round2.json`). None of these searches covered agent-executed reproduction outside cosmology and HEP; the round-3 novelty referee’s sweep without the word “BAO” found three such pipelines on its first page. For this revision we ran one recorded sweep on that axis and read those three papers and three further ones by arXiv id (`literature_search_log_round3.json`). Only papers we read are

Table 6: Derived quantities.

Quantity	Value	Definition / source key
DR1→DR2 shift $\Delta\Omega_m$ (DR2 run, emcee means)	0.00283	DR1_to_DR2_shift.Delta_0m
$\Delta\Omega_m/\sigma_{\text{nested}}$ (own $\sigma_{\text{nested}} = 0.01194$)	0.237	$\sigma_{\text{nested}}^2 = \sigma_{\text{DR1}}^2 - \sigma_{\text{DR2}}^2$; Delta_over_sigma_nested_own
$\Delta\Omega_m/\sigma_{\text{nested}}$ (preregistered $\sigma = 0.01229$)	0.231	Delta_over_sigma_nested_prereg
$\Delta\Omega_m/\sigma_{\text{indep}}$ (wrong if nested; for comparison)	0.167	Delta_over_sigma_independent_own
$\Delta\Omega_m/ \sigma_{\text{DR1}} - \sigma_{\text{DR2}} $ ($C = 1$ bracket)	0.46	round1_response_checks.json: dr1_dr2_conventions.ratio_C1
SDSS-DESI DR2 tension, $(\Omega_m, h r_d)$, 2 dof	$N_\sigma = 0.877$	$\chi^2 = 1.93$, PTE = 0.381; tension.primary_emcee_moments
same, grid / MAP+Laplace / KDE shift	0.862 / 0.932 / 0.939	tension.{grid_moments, MAP_laplace, posterior_parameter_shift_KDE}
same, with $X + X^T$, $X = \rho\sqrt{C_S}\sqrt{C_D}$, subtracted, $\rho = 0.57$ (illustration)	$N_\sigma = 1.34$	round1_response_checks.json: overlap_sensitivity
H_0 shift, exact CAMB r_d minus fitting formula, DR2, MAP (deterministic; equals the propagated measured formula error, which exceeds Aubourg’s stated 0.021%; not new)	−0.0419 km/s/Mpc	round3_response_checks.json, key formula_shift, map_diff
same, DR1, MAP	−0.0411 km/s/Mpc	...T2_DR1.map_diff
posterior-mean differences DR2 / DR1 (MC-noise dominated)	-0.034 ± 0.015 / -0.0044 ± 0.020	measured_formula_systematic; MC error from chain std/ $\sqrt{\text{ESS}}$
Wrong neutrino bookkeeping (N4), ΔH_0 (MAP)	0.164 km/s/Mpc	$\Delta\chi^2 = -5.2 \times 10^{-5}$; controls/N4.json

Table 7: Dual-environment Lean check (`results/cosmo_synthesis/lean_dual_check.json`, revision 2 of the checker). “#print axioms” counts commands in comment-stripped source. Wall times are single measurements on a shared machine.

File	Theorems	#print axioms	Env 1 rc	Env 1 time [s]	Env 2 rc	Env 2 time [s]
BA0_FlatLCDM.lean	3	3	0	16.0	0	645.0
DESI_DR2_wCDM.lean	6	6	0	108.2	1 (BLOCKED_ENV)	138.0
BA0_BBN_H0.lean	10	10	0	23.2	0	367.6
BA0_Consistency.lean	9	9	0	10.6	0	313.1

cited.

For agent-executed reproduction of a published result against numeric targets, the most direct precedent is SHARP [30]: a Claude Code pipeline (claude-opus-4.6) whose initiating prompt names the paper and the target metrics, with Paper Analyst, Code, Test, Statistician and Critic sub-agents and human review at checkpoints; it reports three independent runs against the published ParticleNet-Lite numbers, validates checkpoints with an external evaluation script written by a human expert, and records a truth-label-leakage failure that “went undetected by automated tests”, the class of problem our N4 control targets. At scale, Huang [31] runs a fresh Claude Code/Opus 4.6 agent on each of 111 open-access Quantum ESPRESSO papers, with a four-tier verdict (T1–T4) and a 12-paper two-machine cross-check with no shared state, and matches 75.8% of 571 claims within 5% of the published value. A companion pipeline [32] reproduces $k = 5$ published anchors before any new computation and uses fresh-context adversarial review “prompted to find rather than confirm”; its no-pilot ablation had the anchor value in its own files yet never wrote the comparison, the failure mode that preregistered targets and tolerances address, and its note added reports that the anchor value itself was not converged, so that single-anchor calibration gives “consistency with published literature, not truth”. Our targets (DESI posterior means, and our unexplained DR1 χ^2 offset) sit in the same epistemic position. For cosmology, the most direct precedent is cmbagent. Its 2024 workshop version [9] reproduced the published ACT DR6 lensing constraints with a human approving each step, and its authors note that such output cannot be confirmed without redoing

Table 8: Elenchus ledgers. “A audited (model)” counts Tier A rows whose audit is a non-empty object naming an auditor (all are model referees). “Audit hash = current file” counts those audits whose recorded source sha256 equals the current Lean file. rc is the gate exit code (DR1: from the DR2 run’s parent-ledger control). Tier B here means a seeded floating-point harness on sha256-verified inputs; Elenchus’s own Tier B is exact $\mathbb{Q}/\mathbb{Z}$ arithmetic and would file these rows at X (Sec. 3.5).

Run	Claims	A	B	L	C	A audited (model)	audit hash = current file	gate rc
DR1 flat- Λ CDM	6	3	1	2	0	0/3	0/0	1
DESI DR2	25	6	8	10	1	6/6	6/6	0
BAO+BBN H_0	43	10	17	15	1	8/10	0/8	1
eBOSS vs DESI	27	9	8	8	2	9/9	0/9	0

the analysis independently. Its Planning & Control mode [17] runs “with no human-in-the-loop at any point” and serves as the analysis backend of Denario [18], an end-to-end research-assistant system. Related systems include agent-driven inference pipelines in which human redirection was decisive [10] (a competition entry, not a reproduction of a published result), agentic cosmology frameworks and their caveats [11], and DarkAgents [12], which audits assumptions and compares its results against a human analysis. The AI Cosmologist [28] is an autonomous agentic system for cosmological and astronomical data analysis, demonstrated on machine-learning tasks (Galaxy Zoo 2, Quijote) rather than on reproductions of published parameter constraints. Gandhi, Bolliet & Zubeldia [29] extend cmbagent with plots treated as checkpoints that a vision-language model judges against generated rubrics, which also leaves auditable reasoning traces. The nearest precedents for a model referee are cosmology-native: Denario’s Review module writes a referee report (`referee.md`) [18, Sec. 3.7], and the VLM judge of [29]; the AI Scientist [19] also uses an automated reviewer. The closest architectural precedent for our design is JFC (“Just Furnish Context”, Moreno et al. [27]): a Claude Code orchestrator delegates to executor and reviewer subagents, “multi-agent review replaces the human feedback loop”, “human oversight is concentrated at a single formal unblinding gate”, an append-only experiment log records every phase transition, and one of its analyses is checked against, and consistent with, a published CMS $H \rightarrow \tau\tau$ measurement. In the pages of JFC and cmbagent that we read we did not find preregistered numeric targets or hash-locked results; we do not claim that they lack them. SHARP’s prompt does fix numeric targets, without tolerances, controls or a lock; that, rather than the targets themselves, is what differs here. Benchmark results on reproduction are mixed. Benchmarks built from whole papers report low scores: ReplicationBench [13] reports that the best model, Claude 4.5 Sonnet, scores 22% (Sec. 5.1 and Table 2; its abstract says “under 20%”), and PRBench [14] reports a best overall score of 34%, a zero end-to-end success rate, and fabricated outputs. Pipelines working from public inputs with a stated method report high agreement (SHARP; 75.8% within 5% in [31]), as do our four runs, which start from public summary likelihoods with a published estimator. Collider-Bench [20] has agents reproduce LHC analyses and uses an LLM provenance judge that flags FABRICATED runs whose values cannot be traced to an executed pipeline; this is the closest analogue of our re-running referee plus content-addressed evidence. On the formal side, HepLean (now Physlib) [15, 16] formalizes high-energy physics in Lean 4, a Lean formalization found a genuine error in a widely cited 2HDM stability theorem [16], and FormalScience [21] autoformalizes physics in Lean with agents and gives a taxonomy of semantic drift (e.g. “abstraction elevation”), the class of problem behind our DR2 $z > -1 \rightarrow z \geq 0$ drift.

Blind analysis. The field’s established safeguard against confirmation bias is blinding. Both

target analyses were blind: DESI DR1 blinded catalogs (redshift shifts for galaxies and quasars, an additive BAO-peak shift for $\text{Ly}\alpha$) until a prescribed set of validation tests passed [1, Sec. 2.3.1], with the scheme validated by Andrade et al. [22], and DR2 applied “strict catalog-level blinding” with pre-specified unblinding criteria [2, Secs. II.A.1 and IX]. Grosso, Mikuni & Heinrich [23] recommend a blind-analysis-like separation for agentic systems (optimize on simulation, reserve observed data for the final test) and note that “No analogous verification infrastructure exists for physics analyses” to the role Lean plays for mathematics. Our setup is *weaker* than a blind analysis: the targets were published and visible to the agents, and the preregistrations were committed to git only after the fits. Preregistration and the hash-locked unread fit guard the decision rules, not the agents’ knowledge of the answer. It is also weaker than JFC [27] in this respect: JFC blinds its agents in stages (Asimov data, then a 10% subsample, then the full data) and requires a human to approve unblinding, whereas our pipeline has no blinding and no human gate at any point. The published cosmology instance of our hash-lock safeguard is TOPO [26]: the analyst signs an Analysis-Hash of code commits and input files and publishes it on a timestamped medium (their example uses a public blockchain) before running, then publishes Merkle-tree roots over the time-ordered MCMC chain, so that a verifier can re-run part of the chain and check it; it is demonstrated with Cobaya and CLASS on BAO likelihoods that include SDSS DR16. Our hash-locked unread fit and sha256 evidence blobs are a one-file instance of that idea, timestamped only by file mtime, and TOPO’s timestamped commitment is exactly what our preregistrations lack.

What is new, within the limits of that search.

1. *The combination of safeguards applied to agent-executed cosmology:*

- preregistered targets and tolerances;
- an amendment that hash-locks an unread earlier result and reports both analyses (a local version, timestamped only by file mtime, of the commitment that TOPO [26] timestamps publicly);
- a subtle-bug control that the likelihood cannot detect;
- in-file, axiom-audited Lean statements of the model skeleton, checked in two environments (same toolchain) with `sorry` and axiom-smuggling controls;
- a tier-capped claim ledger with mutation controls;
- a re-running adversarial referee;
- a second-language port.

Each item exists elsewhere, the closest combined precedents being JFC [27] and, for reproduction against numeric targets, SHARP [30], and our combination is weaker than the blind analyses of the target papers and than JFC’s staged blinding and human gate in one respect (the answer was visible, and no person approved anything). We did not find these items combined in the literature we searched, and we do not claim priority beyond that.

2. *Two derived consistency numbers* not quoted in the fetched text of [2, 1]: the DR1→DR2 shift in nested σ , and the BAO-only ($\Omega_m, h r_d$) SDSS–DESI DR2 N_σ from eq. (18). Each is a simple function of public inputs.
3. *Verification artifacts and an audit of the process*, including where it failed: a model audit clears the ledger gate, and the gate does not check claim-to-evidence correspondence; the project’s Lean runner could not gate worktree files; preregistration files were untracked; one negative control was vacuous; one Lean statement drifted ($z > -1 \rightarrow z \geq 0$) and stays

drifted in the shipped file for audit binding, although the preregistered domain was shown to prove with the same tactics for the two positivity theorems; the eBOSS preregistered module `BA0_TensionStatistic.lean` (diagonal χ^2) shipped as `BA0_Consistency.lean` (full 2×2 covariance), a deviation disclosed in the eBOSS README and paper; the H0 script passes gate G1 where the preregistration names G2; the H0 statement audits are bound to a stale source hash; the first dual-check record was hand-corrected; the primary H0 DR1 verdict flips from PARTIAL to PASS when the frozen script is re-run under the project’s other Python environment, and no run recorded its interpreter; the ledgers’ Tier B is a local redefinition of Elenchus’s exact-arithmetic tier; and defects were found in a Rust reimplement of CVODE.

6 Limitations

- **Partial result, environment-dependent.** The primary H0 DR1 check is PARTIAL ($|\Delta H_0| = 0.167$) in the recorded run (`venv-pt`) and PASS (0.1455) when the same frozen script is re-executed under `venv-cosmo`; the two differ by 1.5 Monte Carlo errors, so the strict boundary is not resolved and neither label is a stable property of the analysis. The secondary DR1 check is PARTIAL in both. The gate-tolerance mechanism is consistent with the recorded PARTIAL but does not establish it: depending on which DR1 hr_d is the reference and on its rounding, the mechanism accounts for $0.05\text{--}0.25 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (± 0.038) of the observed 0.167, and the explanation is post-hoc. The SDSS hr_d target comes from a secondary source.
- **Incoherent H0 tolerances.** The preregistered gate tolerance (0.25σ on hr_d) admits H_0 offsets of up to $0.25 / 0.45 \text{ km s}^{-1} \text{ Mpc}^{-1}$, beyond the strict window, so PASS versus PARTIAL does not isolate the BBN/ r_d step. A published BBN reaction-network systematic of up to $\sim 0.5\sigma$ in h [33] is also larger than the strict window. No tolerance was changed after the fits. The H0 script uses gate G1 in the T2 verdict where the preregistration names G2 (G2 passed; the verdict is unchanged; the frozen script was not edited).
- **DR1 χ^2 offset.** Our DR1 $\chi^2_{\min}$ exceeds DESI’s by 0.081 on the same data; radiation and neutrino background choices do not explain it. It remains open.
- **Tier A is model-audited, by the same model family.** Every statement audit was done by a Claude model reviewing Claude-written work, so producer and verifier are not independent in vendor or training lineage, and automated review is reviewer-family dependent [35]. Two H0 theorems are unaudited. No person has checked that any Lean statement means what its gloss claims. The Lean theorems are elementary and do not certify any fit; the H0 degeneracy and identifiability theorems have the scope limits stated in Sec. 3.4, and nothing is proved about the CAMB r_d .
- **Ledger binding and gate scope (not fixable here).** The H0 audits are bound to a stale, never-committed source hash, and the H0 ledger does not name the change; the three ledgers use three audit schemas; some computed rows carry kind `citation`; the gate does not check that an evidence blob belongs to its claim, an empty audit object clears the Tier A flag, and a Tier A blob that lists `sorryAx` passes the gate; all 34 Tier B rows are floating-point outputs that Elenchus’s own semantics would file at X, and the DR1/DR2 ledger READMEs do not disclose this. The ledgers (including their `kind` fields and READMEs) and the Elenchus tool are outside this synthesis’ write scope; re-kinding the float rows (e.g. a `float_harness` kind

capped below B, or refiling at X) belongs in the ledgers themselves. A standard audit schema with `auditor_kind`, the audited and current source hashes and the change since audit is needed in the ledgers themselves.

- **Lean statements not changed.** The DR2 domain drift ($z \geq 0$ instead of the preregistered $z > -1$) and the inert $z \neq -1$ hypothesis of `distance_duality` remain in the committed files, which are outside this synthesis’ write scope. For `E2_pos` and `E_pos` the preregistered domain was tested and proves with the same tactics; the other DR2 theorems were not tested on $z > -1$, for which the preregistration fixed no domain.
- **Lean gate deviation.** `lean_runner.py` was not used. Checks used `lake env lean` with in-file `#print axioms`. One file could not be checked in environment 2, and both environments share the Lean binary and Mathlib commit. The three new modules are not yet part of the project’s `lake build`.
- **Not a blind analysis.** The published targets were visible to the agents, and preregistrations were committed only after the fits.
- **Preregistration provenance.** The preregistration files were not committed before fitting, so mtime is the only evidence of ordering. In the eBOSS file, the amendment field’s timestamp is later than the file’s mtime. This is disclosed in its ledger README. The fix is a timestamped commitment made before the fit (for example a signed hash on a public timestamp service, as in TOPO [26]), or at least a git commit pushed before the fit.
- **Radiation and neutrinos.** In the BAO-only runs, radiation is off in the primary model and the massive neutrino is not modelled. The sensitivity shifts are small (-0.046σ and 0.032σ for DR2) but not zero.
- **Overlap covariance unknown.** The SDSS–DESI statistic assumes independent surveys. Because the surveys overlap, N_σ is a lower bound (and the PTE an upper bound) only if the parameter-level cross-covariance is positive; its size is model-dependent ($N_\sigma = 1.34$ for the illustrative $\rho = 0.57$, which is DR2’s value for one bin pair). At the distance level DR2 finds $1.9\text{--}2.6\sigma$ for the LRG2/eBOSS-LRG bin pair.
- **Controls.** The SDSS-baseline EdS control was vacuous as first coded. The fix gates it on the Gaussian summary instead. The eBOSS injection test does not exercise the SDSS baseline grid likelihood. The scrambled-covariance control discriminates only in aggregate. N4 is post-hoc and not part of the H0 verdict.
- **Estimators.** The DR1 run and the Rust port compare χ^2 minima against DESI posterior means; for DR1 the matched best-fit comparison is also given (Sec. 4.1).
- **Project completion criteria not met.** The recorded logs of the three preregistered runs give: `antigravity_guard.py` FAIL / FAIL / FAIL (DR2 / H0 / eBOSS), with findings that include missing optional packages and, for eBOSS, `camb` outside `venv-cosmo`; `test_rigor_guard.py` PASS / PASS / PASS; and `pytest tests/`: DR2 “22 failed, 1358 passed, 49 skipped, 22 warnings, 8 errors”, H0 collection interrupted (“1 warning, 21 errors”), eBOSS “22 failed, 1358 passed, 49 skipped, 22 warnings, 8 errors”. The eBOSS referee found that none of the failing tests mentions that problem. We therefore do not claim that the repository’s completion checks pass. `pytest` was not re-run for this synthesis.

- **Solver.** The CVODE measurements quoted from the Rust README predate the PR #60 fix and were not repeated.
- **Unrecorded numerical environment (not fixable in the run files).** No result JSON records its interpreter or library versions. The environment of each recorded run is inferred from re-execution (Reproducibility). The note in `results/bao_bbn_h0/fit.json` that calls the secondary-versus-locked identity “expected”, and the docstring of the DR1 fit script (`fit_desi_bao.py`), which names `venv-pta` although the script fails there, are outside this synthesis’ write scope and were not changed.
- **Single execution per problem.** Each problem was run once by the agent pipeline; there are no repeat agent runs and no cross-model runs.

7 Reproducibility

Code lives under `scripts/{bao_flcdm, desi_dr2_bao, bao_bbn_h0, eboss_vs_desi}/`, results under `results/<run>/`, Lean files under `formal/ANSE/`, and this paper under `papers/cosmo_synthesis/`. Two Python environments were used, and the H0 result depends on which (Sec. 4.3): `PTA=/mnt/disks/disk-socrateai-local-1/venv-pta/bin/python` (Python 3.10.12, numpy 1.26.4, scipy 1.15.3, no CAMB) and `COSMO=/mnt/disks/disk-socrateai-local-1/venv-cosmo/bin/python` (Python 3.11.16, numpy 2.4.6, scipy 1.17.1, CAMB, CLASS), both with emcee 3.1.6. No run JSON records its interpreter; the environment of each recorded result was established by re-execution. DR1 fit: COSMO (re-run reproduces the JSON leaf for leaf apart from its timestamp; the script cannot start under PTA, whose Python lacks `datetime.UTC`, although its docstring names PTA). H0 fit: PTA (0 differing leaves; 751 under COSMO). DR2 chains: bit-identical under both (round-3 referee), so the interpreter cannot be determined from the outputs; the scripts’ docstrings and the DR2 preregistration’s `parent_import` note name PTA. eBOSS: PTA (the round-3 referee reproduced its positive control leaf for leaf there). The CAMB table and the CAMB cross-checks need COSMO. Synthesis scripts: COSMO. Key commands, with the interpreter used for the recorded outputs (run from the repository root unless a `cd` is shown):

```
COSMO scripts/bao_flcdm/fit_desi_bao.py
PTA scripts/desi_dr2_bao/controls.py; PTA scripts/desi_dr2_bao/mcmc_fit.py
PTA scripts/desi_dr2_bao/grid_posterior.py; PTA scripts/desi_dr2_bao/assemble_fit.py
COSMO scripts/bao_bbn_h0/rd_camb_table.py; PTA scripts/bao_bbn_h0/fit_real.py
cd scripts/bao_bbn_h0 && PTA control_n4.py run # interpreter as named in its docstring
PTA scripts/eboss_vs_desi/fit_eboss_vs_desi.py; PTA scripts/eboss_vs_desi/negative_controls.py
COSMO results/cosmo_synthesis/run_lean_dual_check.py # no numerics; drives lake env lean
COSMO results/cosmo_synthesis/dr1_chi2_offset_check.py
COSMO results/cosmo_synthesis/round1_response_checks.py
COSMO results/cosmo_synthesis/round2_response_checks.py
COSMO results/cosmo_synthesis/round3_env_rerun_setup.py # then fit_real.py in /tmp/r3resp/{cosmo,
pta} under each
COSMO results/cosmo_synthesis/round3_response_checks.py
COSMO results/cosmo_synthesis/round3_m6_sorry_blob_mutation.py
COSMO results/cosmo_synthesis/extract_provenance.py
cd papers/cosmo_synthesis && COSMO gen_numbers.py && COSMO make_figures.py
pdflatex cosmo_synthesis.tex; pdflatex cosmo_synthesis.tex
cd rusty-SUNDIALS && cargo test -p qf-bao-distances -- --nocapture
```

Data: DESI DR1/DR2 Gaussian BAO files (Table 9), attributed to [1, 2]; SDSS DR16 BAO likelihoods from [3, 4], with sha256 hashes pinned in `results/eboss_vs_desi/preregistration.json`.

Planck 2018 [6] supplies the CAMB anchor, and the BBN prior comes from [7]. Table 10 lists the hashes of the key JSONs. For each preregistration, it also checks the hash against the value pinned inside the corresponding fit.

Table 9: DESI data files.

DESI data file (as used by all DESI fits)	sha256
desi_gaussian_bao_ALL_GCcomb_mean.txt	9ac154ab583ce759c0f7eef3c978c7c70a6ead2d18774caceadf1a350a640585
desi_gaussian_bao_ALL_GCcomb_cov.txt	252a143274c8a07c78694c119617d36594f6d7965d00319ca611c6ffb886e509
desi_2024_gaussian_bao_ALL_GCcomb_mean.txt	dd2873a0b88459a491af3c0c0307ba059f62d6f9211d5b976760f310565a1be68
desi_2024_gaussian_bao_ALL_GCcomb_cov.txt	bbafa9074b51cf1a45e0d10e4f37db8c0e80a5d1d1788857abb7fc49fb21abcc

Table 10: sha256 of key JSONs, computed by `gen_numbers.py` at build time.

File	sha256	Matches pin in fit
results/bao_flcdm/desi_dr1_flcdm_fit.json	b6819ad6adfe00c4408b64b83e50d10b219d6b506853a86e54433256fee5de64	
results/desi_dr2_bao/preregistration.json	723cd27d3e5e198eb415ce3f585c4dfcdd3fa1af658660c8b764bf3476fd32a4	yes
results/desi_dr2_bao/fit.json	4380af6d0e1ed743a2981c517e7fff7e22a49eb1e2952a7788918da7daeeae95	
results/desi_dr2_bao/controls.json	f18dc16e49722db7ff64a1045a8075a5838856ddb94ad8ad073d8b69c56d5119	
results/bao_bbn_h0/preregistration.json	5bcd156f38c2218867aa43eb69840033dbdeef504258cbbdda0fd345f3bbb12	yes
results/bao_bbn_h0/preregistration_amendment_1.json	b0c7b80d885bf15a9becf0c3f9b3cae0419b9abaef3c9f7dd0ed840d2265af64	yes
results/bao_bbn_h0/fit_attempt1_fittingformula_UNREAD_AT_AMENDMENT.json	b7fe5608e4ae6d0b1cb0f7844e55a6025089336cc9bcb3086e9ead6e85531910	
results/bao_bbn_h0/fit.json	8974d5d29ad7bb5b3a71ec336b3205452493aa05eede328ce0bac01cd557d3aa	
results/eboss_vs_desi/preregistration.json	e0eca89f279cfaac3a8ad28adf06a09a3805e2a36d907f7d058482b5e6822f8a	yes
results/eboss_vs_desi/fit.json	23f0d1d10062ba8152b8546510ffe9e0114d6809182e2e6d35f583d147c5628e	
results/cosmo_synthesis/lean_dual_check.json	af6242e2eccdd63e407d118cd4799fc9a4364792ab0035018eee78b778a287e5	
results/cosmo_synthesis/rust_crosscheck.json	75c67cc2aa9ca84b88e4e130fb5b67aed735e58936bd80d59e873a155d8ee5fa	
results/cosmo_synthesis/model_tier_provenance.json	8cc513fe45159459242da2a4908d3cc73400208a9ea3599136e556427462c34	
results/cosmo_synthesis/round1_response_checks.json	c0715a9ae6b82654b1bad2c7efcb88fcd1ccdde244556039ad7d732183ef1611	
results/cosmo_synthesis/dr1_chi2_offset_check.json	adb517119a02fa794ea67305ba18bc80b8cd9495d74b64118f16b0451ec3977f	
results/cosmo_synthesis/lean_dual_check_v1_handedited.json	bc09047fb60173ec903d3cc7c1f0a326e848da0d01e445e97ef75e3f6841dd430	
results/cosmo_synthesis/round2_response_checks.json	cd5f056c00d77afa4b4d3254449e7f16aa5c8882bef080287a131a7e9b13847a	
results/cosmo_synthesis/dr2_zgtm1_check.lean	ca31d646d0500ff5a29b58a6c2d88ef5f07f7eb994b4c52878d13b0f56cb01c7	
results/cosmo_synthesis/round3_response_checks.json	c7dfd1e5775cf150445e83025a118f13da6c5d0111c5a9fb0c1d511318bd6f45	
results/cosmo_synthesis/round3_m6_mutation.json	d80e072c237e1ff2e1009633cdac3c448d4236eb20919872914f7f1cf34168bc	
results/cosmo_synthesis/round3_rerun_fit_venv_pta.json	620a03cad70c25d2f50c48095f1d7e453d93e34d5e0c01c3b12ad4705ed1cb49	
results/cosmo_synthesis/round3_rerun_fit_venv_cosmo.json	3dfcd9e27c729c2470f700e0e434d33c32571e8926d68dfd7609b0da5d29b24e	

References

- [1] DESI Collaboration, *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, arXiv:2404.03002.
- [2] DESI Collaboration, *DESI DR2 Results II: measurements of baryon acoustic oscillations and cosmological constraints*, arXiv:2503.14738.
- [3] S. Alam et al. (eBOSS), *Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey: cosmological implications from two decades of spectroscopic surveys at the Apache Point Observatory*, arXiv:2007.08991.
- [4] H. du Mas des Bourboux et al., *The Completed SDSS-IV Extended Baryon Oscillation Spectroscopic Survey: Baryon Acoustic Oscillations with Lyman- α Forests*, arXiv:2007.08995.
- [5] É. Aubourg et al., *Cosmological implications of baryon acoustic oscillation measurements*, arXiv:1411.1074.

- [6] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, arXiv:1807.06209.
- [7] N. Schöneberg, *The 2024 BBN baryon abundance update*, arXiv:2401.15054.
- [8] X. Callens, *SocrateAI-Scientific-Elenchus* (software; the author’s own, unpublished, not peer-reviewed), github.com/xaviercallens/SocrateAI-Scientific-Elenchus.
- [9] A. Laverick et al., *Multi-Agent System for Cosmological Parameter Analysis*, arXiv:2412.00431.
- [10] T. Borrett et al., *Competing with AI Scientists: Agent-Driven Approach to Astrophysics Research*, arXiv:2604.09621.
- [11] L. Xu and T. Borrett, *Beyond AI as Assistants: Toward Autonomous Discovery in Cosmology*, arXiv:2605.14791.
- [12] M. Lucente, S. Pascoli, F. Sala, M. Zandi, *DarkAgents: towards an agentic system for theoretical astroparticle physics*, arXiv:2606.11157.
- [13] C. Ye et al., *ReplicationBench: Can AI Agents Replicate Astrophysics Research Papers?*, arXiv:2510.24591.
- [14] S. Qiu et al., *PRBench: End-to-end Paper Reproduction in Physics Research*, arXiv:2603.27646.
- [15] J. Tooby-Smith, *HepLean: Digitalising high energy physics*, arXiv:2405.08863.
- [16] J. Tooby-Smith, *Formalizing the stability of the two Higgs doublet model potential into Lean: identifying an error in the literature*, arXiv:2603.08139 (states that PhysLean was renamed Physlib).
- [17] L. Xu et al., *Open Source Planning & Control System with Language Agents for Autonomous Scientific Discovery*, arXiv:2507.07257.
- [18] F. Villaescusa-Navarro, B. Bolliet, P. Villanueva-Domingo et al., *The Denario project: Deep knowledge AI agents for scientific discovery*, arXiv:2510.26887.
- [19] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, D. Ha, *The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery*, arXiv:2408.06292.
- [20] D. A. Faroughy, S. Palacios Schweitzer, I. Pang, S. Mishra-Sharma, D. Shih, *Collider-Bench: Benchmarking AI Agents with Particle Physics Analysis Reproduction*, arXiv:2605.13950.
- [21] J. Meadows, L. Zhang, A. Freitas, *FormalScience: Scalable Human-in-the-Loop Autoformalisation of Science with Agentic Code Generation in Lean*, arXiv:2604.23002.
- [22] U. Andrade et al., *Validating the Galaxy and Quasar Catalog-Level Blinding Scheme for the DESI 2024 analysis*, arXiv:2404.07282.
- [23] G. Grosso, V. Mikuni, L. Heinrich, *Are We Ready for AI-Driven Discovery? AI Verification Before the Next Fundamental Physics Breakthrough*, arXiv:2607.10039.
- [24] B. Ghosh, C. Bengaly, *Consistency tests between SDSS and DESI BAO measurements*, arXiv:2408.04432.
- [25] A. C. Ferri, Ruchika, A. Melchiorri, *Present Day Cosmic Acceleration from SDSS and DESI BAO: A Call for Finer Tomography of the DESI Bright Galaxy Survey*, arXiv:2607.07348.

- [26] S. Casas, C. Fidler, *TOPO: Time-Ordered Provable Outputs*, arXiv:2411.00072.
- [27] E. A. Moreno, S. Bright-Thonney, A. Novak, D. Garcia, Y. Zhao, P. Harris, *AI Agents Can Already Autonomously Perform Experimental High Energy Physics*, arXiv:2603.20179.
- [28] A. Moss, *The AI Cosmologist I: An Agentic System for Automated Data Analysis*, arXiv:2504.03424.
- [29] K. Gandhi, B. Bolliet, I. Zubeldia, *Enhancing Agentic Autonomous Scientific Discovery with Vision-Language Model Capabilities*, arXiv:2511.14631.
- [30] J. Birk, G. Kasieczka, S. Mishra-Sharma, B. Nachman, D. Noll, T. Wamorkar, *A Scientific Human-Agent Reproduction Pipeline*, arXiv:2604.18752.
- [31] H. Huang, *Grounded autonomous scrutiny at scale: emergent critique from reproduction of published computational physics papers*, arXiv:2604.12198.
- [32] H. Huang, *Grounded autonomous research: a fault-tolerant LLM pipeline from corpus to manuscript in frontier computational physics*, arXiv:2607.02329.
- [33] M. Akharman, N. DePorzio, C. Giovanetti, H. Liu, *The Role of Big Bang Nucleosynthesis in Joint Cosmological Analyses*, arXiv:2609.12065.
- [34] J. Kim, D. F. Mota, A. Tamosiunas, *A Sequentially-Valid Reanalysis of DESI's Dynamical Dark Energy Signal*, arXiv:2607.28918.
- [35] V. L. Ravideshik, M. Kejriwal, *Can AI Evaluate AI Scientists? A Benchmarking Study of Autonomous Research Generation Systems Using Automated Multi-Model Review*, arXiv:2607.28631.

A Lean statements (verbatim) and axiom output

The statements below were extracted by `gen_numbers.py` from the committed files. They are verbatim up to the typesetting of Unicode symbols (for example $\mathbb{R}$, $\leq$, f). The script checks that the appendix holds exactly 28 theorem blocks, whose names match the declarations in the dual-check axiom output. The file hashes match those recorded by the dual check; the script aborts if they do not. The environment-1 `#print axioms` output for all 28 theorems follows the statements.

```
formal/ANSE/BA0_FlatLCDM.lean
```

```
Namespace ANSE.BA0FlatLCDM; source sha256
```

```
c85c1e8abe28b2652efb2a35f91399034087450b8c7af4dc308530d72badd702.
```

```
Proof bodies replaced by := ...; statements and definitions verbatim.
```

```
noncomputable def E (Om z : ℝ) : ℝ := Real.sqrt (Om * (1 + z) ^ 3 + (1 - Om))
```

```
theorem E_pos {Om z : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) (hz : 0 ≤ z) :  
  0 < E Om z := ...
```

```
theorem E_strictMonoOn {Om : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) :  
  StrictMonoOn (E Om) (Set.Ici (0 : ℝ)) := ...
```

```
noncomputable def D_L (z Dc : ℝ) : ℝ := (1 + z) * Dc
```

```

noncomputable def D_A (z Dc : ℝ) : ℝ := Dc / (1 + z)

theorem distance_duality {z Dc : ℝ} (hz : z ≠ -1) :
  D_L z Dc = (1 + z) ^ 2 * D_A z Dc := ...

formal/ANSE/DESI_DR2_wCDM.lean

Namespace ANSE.DESIDR2wCDM; source sha256
3b451bd5a93d682cd05c99fad4681808cb5c18877aa6389dda545dbace5e5adc.
Proof bodies replaced by := ...; statements and definitions verbatim.

noncomputable def E2 (Om w z : ℝ) : ℝ :=
  Om * (1 + z) ^ 3 + (1 - Om) * (1 + z) ^ (3 * (1 + w))

noncomputable def E (Om w z : ℝ) : ℝ := Real.sqrt (E2 Om w z)

noncomputable def Dc (Om w z : ℝ) : ℝ := ∫ t in (0:ℝ)..z, (E Om w t)-1

theorem E2_pos {Om w z : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) (hz : 0 ≤ z) :
  0 < E2 Om w z := ...

theorem E_pos {Om w z : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) (hz : 0 ≤ z) :
  0 < E Om w z := ...

theorem E_w_neg_one (Om z : ℝ) : E Om (-1) z = ANSE.BAOFlatLCDM.E Om z := ...

theorem E_strictMonoOn_of_neg_one_le {Om w : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1)
  (hw : -1 ≤ w) : StrictMonoOn (E Om w) (Set.Ici (0 : ℝ)) := ...

theorem invE_continuousOn {Om w : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) :
  ContinuousOn (fun t => (E Om w t)-1) (Set.Ici (0 : ℝ)) := ...

theorem Dc_strictMonoOn {Om w : ℝ} (hOm0 : 0 < Om) (hOm1 : Om < 1) :
  StrictMonoOn (Dc Om w) (Set.Ici (0 : ℝ)) := ...

formal/ANSE/BAO_BBN_H0.lean

Namespace ANSE.BAOBBNH0; source sha256
2b6530838efa0df9bfb475a5d8b25ee29aa121989cc83d04e07f595f5f616e38.
Proof bodies replaced by := ...; statements and definitions verbatim.

noncomputable def omegaNu : ℝ := 0.0107 * 0.06

noncomputable def rdAubourg (wcb wb : ℝ) : ℝ :=
  55.154 * Real.exp (-72.3 * (omegaNu + 0.0006) ^ 2) /
  (wcb ^ (0.25351 : ℝ) * wb ^ (0.12807 : ℝ))

noncomputable def omegaCb (h Om : ℝ) : ℝ := Om * h ^ 2 - omegaNu

theorem rd_numerator_pos : 0 < 55.154 * Real.exp (-72.3 * (omegaNu + 0.0006) ^ 2) := ...

theorem rdAubourg_pos {wcb wb : ℝ} (hcb : 0 < wcb) (hb : 0 < wb) :
  0 < rdAubourg wcb wb := ...

theorem rdAubourg_strictAntiOn_cb {wb : ℝ} (hb : 0 < wb) :
  StrictAntiOn (fun wcb => rdAubourg wcb wb) (Set.Ioi (0 : ℝ)) := ...

```

```

theorem rdAubourg_strictAntiOn_b {wcb : ℝ} (hcb : 0 < wcb) :
  StrictAntiOn (rdAubourg wcb) (Set.Ioi (0 : ℝ)) := ...

theorem rd_strictAntiOn_0m {h wb : ℝ} (hh : 0 < h) (hb : 0 < wb) :
  StrictAntiOn (fun 0m => rdAubourg (omegaCb h 0m) wb) {0m : ℝ | 0 < omegaCb h 0m} := ...

noncomputable def H0 (h : ℝ) : ℝ := 100 * h

noncomputable def DH_over_rd (c h E rd : ℝ) : ℝ := c / H0 h / E / rd

theorem DH_over_rd_eq_hrd (c h E rd : ℝ) :
  DH_over_rd c h E rd = c / (100 * (h * rd) * E) := ...

theorem DH_over_rd_degenerate {h rd h' rd' : ℝ} (c E : ℝ) (hprod : h * rd = h' * rd') :
  DH_over_rd c h E rd = DH_over_rd c h' E rd' := ...

theorem gate_reparam_exact {hrad : ℝ} (hne : hrad ≠ 0) (c E hrd : ℝ) :
  DH_over_rd c hrad E (hrd / hrad) = c / (100 * hrd * E) := ...

theorem hrd_mono_generic {a w 0m h1 h2 : ℝ} (ha : 0 < a) (ha3 : a ≤ 1 / 3) (hw : 0 < w)
  (h0m : 0 < 0m) (hh1 : 0 < h1) (h12 : h1 < h2) (hsupp : w ≤ (1 - 2 * a) * (0m * h1 ^ 2)) :
  h1 * (0m * h2 ^ 2 - w) ^ a < h2 * (0m * h1 ^ 2 - w) ^ a := ...

theorem hrd_strictMonoOn_h {0m wb : ℝ} (h0m : 0 < 0m) (hb : 0 < wb) :
  StrictMonoOn (fun h => h * rdAubourg (omegaCb h 0m) wb)
  {h : ℝ | 0 < h ∧ omegaNu ≤ (1 - 2 * 0.25351) * (0m * h ^ 2)} := ...

```

formal/ANSE/BAO_Consistency.lean

Namespace ANSE.BAOConsistency; source sha256

560e3b5d8266a2e2286f17bc9fc4846853cb927c0259b231d403cef8dd343d8b.

Proof bodies replaced by := ...; statements and definitions verbatim.

```

noncomputable def tension1D (μ1 σ1 μ2 σ2 : ℝ) : ℝ := (μ1 - μ2) ^ 2 / (σ1 ^ 2 + σ2 ^ 2)

theorem tension1D_symm (μ1 σ1 μ2 σ2 : ℝ) :
  tension1D μ1 σ1 μ2 σ2 = tension1D μ2 σ2 μ1 σ1 := ...

theorem tension1D_nonneg (μ1 σ1 μ2 σ2 : ℝ) : 0 ≤ tension1D μ1 σ1 μ2 σ2 := ...

theorem tension1D_eq_zero_iff {μ1 σ1 μ2 σ2 : ℝ} (hσ : 0 < σ1) :
  tension1D μ1 σ1 μ2 σ2 = 0 ↔ μ1 = μ2 := ...

noncomputable def chi2 (a b c d1 d2 : ℝ) : ℝ :=
  (c * d1 ^ 2 - 2 * b * d1 * d2 + a * d2 ^ 2) / (a * c - b ^ 2)

theorem chi2_eq_dotProduct_inv_mulVec (a b c d1 d2 : ℝ) (hdet : a * c - b ^ 2 ≠ 0) :
  ![d1, d2] ·v ((![a, b; b, c])-1 ·v ![d1, d2]) = chi2 a b c d1 d2 := ...

theorem chi2_nonneg {a b c : ℝ} (ha : 0 < a) (hdet : 0 < a * c - b ^ 2) (d1 d2 : ℝ) :
  0 ≤ chi2 a b c d1 d2 := ...

theorem chi2_eq_zero_iff {a b c : ℝ} (ha : 0 < a) (hdet : 0 < a * c - b ^ 2) (d1 d2 : ℝ) :
  chi2 a b c d1 d2 = 0 ↔ d1 = 0 ∧ d2 = 0 := ...

noncomputable def surveyTension (mS1 mS2 aS bS cS mD1 mD2 aD bD cD : ℝ) : ℝ :=
  chi2 (aS + aD) (bS + bD) (cS + cD) (mS1 - mD1) (mS2 - mD2)

```

```

theorem surveyTension_symm (mS1 mS2 aS bS cS mD1 mD2 aD bD cD : ℝ) :
  surveyTension mS1 mS2 aS bS cS mD1 mD2 aD bD cD
    = surveyTension mD1 mD2 aD bD cD mS1 mS2 aS bS cS := ...

theorem posdef_add {aS bS cS aD bD cD : ℝ}
  (haS : 0 < aS) (hdS : 0 < aS * cS - bS ^ 2)
  (haD : 0 < aD) (hdD : 0 < aD * cD - bD ^ 2) :
  0 < aS + aD ∧ 0 < (aS + aD) * (cS + cD) - (bS + bD) ^ 2 := ...

theorem surveyTension_nonneg_and_zero_iff {mS1 mS2 aS bS cS mD1 mD2 aD bD cD : ℝ}
  (haS : 0 < aS) (hdS : 0 < aS * cS - bS ^ 2)
  (haD : 0 < aD) (hdD : 0 < aD * cD - bD ^ 2) :
  0 ≤ surveyTension mS1 mS2 aS bS cS mD1 mD2 aD bD cD ∧
    (surveyTension mS1 mS2 aS bS cS mD1 mD2 aD bD cD = 0 ↔ mS1 = mD1 ∧ mS2 = mD2) := ...

#print axioms output (environment 1)

'ANSE.BA0FlatLCDM.E_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0FlatLCDM.E_strictMonoOn' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0FlatLCDM.distance_duality' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E2_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E_w_neg_one' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.E_strictMonoOn_of_neg_one_le' depends on axioms: [propext, Classical.choice, Quot
  .sound]
'ANSE.DESIDR2wCDM.invE_continuousOn' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.DESIDR2wCDM.Dc_strictMonoOn' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.rd_numerator_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.rdAubourg_pos' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.rdAubourg_strictAntiOn_cb' depends on axioms: [propext, Classical.choice, Quot.sound
  ]
'ANSE.BA0BBNH0.rdAubourg_strictAntiOn_b' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.rd_strictAntiOn_0m' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.DH_over_rd_eq_hrd' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.DH_over_rd_degenerate' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.gate_reparam_exact' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.hrd_mono_generic' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0BBNH0.hrd_strictMonoOn_h' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.tension1D_symm' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.tension1D_nonneg' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.tension1D_eq_zero_iff' depends on axioms: [propext, Classical.choice, Quot.
  sound]
'ANSE.BA0Consistency.chi2_eq_dotProduct_inv_mulVec' depends on axioms: [propext, Classical.choice,
  Quot.sound]
'ANSE.BA0Consistency.chi2_nonneg' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.chi2_eq_zero_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.surveyTension_symm' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.posdef_add' depends on axioms: [propext, Classical.choice, Quot.sound]
'ANSE.BA0Consistency.surveyTension_nonneg_and_zero_iff' depends on axioms: [propext, Classical.
  choice, Quot.sound]

```